

LedgerCrew AI

Product Requirements Document (PRD)

Version 1.0 · 17 September 2026 · Status: In development (hackathon build)

1. Executive Summary

LedgerCrew AI is a multi-agent AI SaaS back-office for Indian micro, small, and medium enterprises (MSMEs). Five specialist AI agents — Intake, Compliance, Collections, Guardrail, and Cash-flow — read invoices, draft GST filings, chase overdue payments, and forecast cash flow, coordinated by an Orchestrator that surfaces a daily business brief. The product targets a hackathon submission first, with a real, monetizable SaaS as the underlying goal.

2. Problem Statement

India has 60M+ MSMEs, nearly all of whom file GST on a recurring schedule and manage invoicing and collections manually. Business owners lose hours every month to three recurring tasks: sorting invoices and receipts for GST compliance, following up with clients who pay late, and estimating whether upcoming cash flow covers near-term obligations like payroll. Existing invoicing tools digitize the paperwork but leave the manual follow-through — compliance checks, collections messaging, forecasting — to the owner.

3. Goals & Objectives

Business goals

- Win the target hackathon by demonstrating a working, judge-appealing, technically credible multi-agent product.
- Lay the foundation for a real, monetizable SaaS (freemium → paid tiers) rather than a throwaway demo.
- Build entirely on free-tier infrastructure through the current development phase.

User goals

- Spend less time on GST paperwork and more time running the business.
- Get paid faster without the discomfort of writing collections messages personally.
- Know, at a glance, whether cash on hand will cover upcoming obligations.

Non-goals (for this phase)

- Full production-grade multi-tenant deployment with paying customers.
- Real WhatsApp Business API integration (a sandbox stands in for now).
- A fully populated cross-business trust score (seed/demo data only until real usage exists).

4. Target Users

Persona	Description	Primary need
Freelancer / solo consultant	Invoices a handful of clients monthly, files GST personally, no accounting staff	Fast invoicing + reminders without hiring help
Small business owner	Runs a shop, studio, or small trading business with a few employees	GST compliance confidence + payroll cash-flow visibility
Growing MSME	10–50 employees, multiple concurrent	Scale collections and compliance

	clients/vendors	without adding back-office headcount
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5. Agent Architecture (Core System)

Agent	Responsibility
Intake	OCR-extracts vendor, amount, date, GST/HSN fields from uploaded invoices/receipts (photo, PDF, or WhatsApp forward)
Compliance	Classifies transactions, flags GST mismatches, drafts a filing-ready GSTR-1/3B return; owns tax-deadline reminders
Collections	Drafts staged payment reminders (polite → firm → final) as invoices age
Guardrail	Reviews every Collections draft before it sends, catching harassment-toned or risky messaging (generator + critic pattern)
Cash-flow	Forecasts 30/60/90-day cash position, flags crunch risk; owns payment-date reminders (e.g. payroll)
Orchestrator	Plans agent execution order, holds shared state, produces the daily Business Brief; reasoning is visibly streamed, not hidden

6. Feature Requirements

Priority key: P0 = required for the core demo/MVP, P1 = strongly desired, P2 = stretch/roadmap.

Feature	Description	Priority	Status
Invoice OCR intake	Extract structured data from uploaded invoice images/PDFs	P0	Backend not built; UI built
GST compliance drafting	Auto-classify + draft GSTR-1/3B filing	P0	Backend not built; UI built
Staged collections reminders	Multi-stage reminder drafting as invoices age	P0	Backend not built; UI built
Guardrail tone review	Second-agent review before any reminder sends	P0	Backend not built; UI built
Cash-flow forecasting	30/60/90-day projection with risk flags	P0	Backend not built; UI built
Live agent reasoning panel	Real-time visibility into what each agent is doing/deciding	P0	UI built (demo data)
Financial calendar	Tax deadlines, escalations, and payment dates in one view	P1	UI built
Ask-your-crew chat	Conversational Q&A over the business's own data	P1	UI built (canned responses)
WhatsApp-native intake	Forward an invoice photo via WhatsApp, zero-onboarding	P1	Not built — planned via Twilio Sandbox
Multilingual voice interface	Ask/answer business questions by voice in Hindi/Bengali/Tamil etc.	P2	Not built
Cross-business trust score	Aggregated, anonymized client payment-reliability score	P2	UI built (seed data only, clearly labeled)
Resource center + GST QnA	Plain-language help plus direct links to official government portals	P1	Built
Privacy policy page	Data handling disclosure incl. sub-processors and DPDP Act rights	P1	Built

7. Non-Functional Requirements

- **Multi-tenancy:** Row-Level Security per business in Supabase from day one; no cross-tenant data leakage.
- **Free-tier operation:** Entire stack (Supabase, Vercel, LangGraph/CrewAI, Groq/Gemini free tiers, Resend, Twilio Sandbox, Groq Whisper, Web Speech API) must run within free-tier limits through the current phase.
- **Resilience:** Every agent has retry/fallback behavior; low-confidence extraction routes to manual review instead of failing silently.
- **Security:** Defense against prompt injection from untrusted uploaded documents; rate-limit handling that degrades gracefully rather than erroring out.
- **Auditability:** Every agent decision is logged; this log doubles as the data source for the reasoning panel and the “show your work” calculation proof.
- **Human control:** A per-business toggle between fully autonomous sends and approve-before-send for Collections output.

8. Design & UX Requirements

Visual language: “ledger + crew” — paper-ledger heritage fused with a mission-control feel for the agent crew. Typography: Fraunces (serif display), IBM Plex Sans (UI body), IBM Plex Mono (tabular data). Color-coding is consistent system-wide: amber = in-progress/needs action, green = done/on-track, red = overdue/escalation. Full token values and component rules are maintained in the separate Design System documentation.

Thirteen pages are planned in total; as of this document, Landing, Login, Dashboard, Ledger, Crew, Privacy, and Resources are built, with Collections partially built (timeline view exists, case-list view does not). Onboarding, Invoice Detail, Compliance/GST Center, Cash-flow deep-dive, Trust Score directory, Settings, and Billing remain to be designed.

9. Success Metrics

Hackathon-stage metrics

- Judge scoring on innovation, technical execution, and market feasibility (per the target hackathon's rubric).
- A working end-to-end demo: upload/forward an invoice → agents process it → visible reasoning → an accurate GST/cash-flow outcome.

Product-stage metrics (post-hackathon)

- Time saved per user per month on invoicing/GST/collections admin (self-reported initially).
- Percentage of invoices with a Collections sequence resulting in faster payment.
- GST filings completed on time via the Compliance draft.
- Free-to-paid conversion rate once monetization is live.

10. Risks & Assumptions

Risk	Mitigation
No real users yet to validate the trust score or real-world accuracy	Dogfooding + outreach to existing audience/community planned before claiming real traction; seed data always clearly labeled as such, never presented as real
Real WhatsApp Business API requires Meta verification + template approval (days, not hours)	Use Twilio's WhatsApp Sandbox for the demo; treat real API onboarding as a post-hackathon task

Free-tier limits (Supabase 500MB DB, LLM rate limits) could be hit under real load	Acceptable at pilot scale (tens of users); a scale-up plan is defined for when limits are approached
Prompt injection via malicious/malformed uploaded documents	Sanitize/validate all extracted fields before they reach downstream agents; never execute raw document text as instructions
Supabase free project auto-pauses after 7 days of inactivity	Scheduled Edge Function heartbeat to keep the project warm during active development/judging

11. Milestones / Phased Plan

- **Phase 1 — Core reliability:** Make the 5-agent loop work end-to-end on real, messy invoice inputs before adding anything new.
- **Phase 2 — Real users:** Acquire a handful of genuine beta users (dogfooding, existing audience, community, personal network) to replace seeded data with real signal.
- **Phase 3 — Hardening:** Prompt-injection defenses, RLS audit, rate-limit fallback handling, a basic automated test suite.
- **Phase 4 — Polish:** Onboarding flow, empty/loading states, a demo video.
- **Phase 5 — Scale story:** A concrete, one-line answer for what happens (technically and financially) beyond free-tier limits.

12. Open Questions

- Which specific hackathon rubric/deadline is this being submitted against, and does that change any priority above?
- What is the target price point and packaging for the paid tier beyond the initial ₹499/month placeholder shown on the landing page?
- Should the cross-business trust score be opt-in or opt-out by default once real users exist?